Combinations vs. Permutations: Subsets & Order

Probability, Random Variables & Combinatorics • Standard: CCSS.MATH.HSS.CP.B.9

BOOK STORY OPENER: Sir Francis Galton's 1873 Bean Machine, The Asteroid Ceres & The Guinness Brewery Chemist

In 1873, Sir Francis Galton built a board studded with pins (the Quincunx) to show how independent random bounces settle into a bell-shaped curve—while in 1908 at the Guinness Brewery in Dublin, chemist William Sealy Gosset ('Student') invented small-sample statistics with n - 1 degrees of freedom! Students roll fair 16 mm polyhedral dice and accumulate trials with [Σ+].

Companion Book: From Blocks to the Stack (SC-6 Ch. 7: Galton's Bean Machine) & Thinking in the Stack (SC-32 Ch. 9: Ceres, Guinness & [Σ

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: Probability Dice & 10-Sector Spinner

Use the 4 precision dice and 10-sector spinner to model multi-stage discrete outcomes.

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! × 5!) = 40320 / (6 × 120) = 56.

3. CALCULATOR ACTION (SC-32 KEYCAPS & STACK TRACE)

Keystroke Flow: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
8	8	Total candidates n = 8
↶ x!	40320	8! = 40,320 in register X
3	3	Subset r = 3 (40320 lifts to Y)
↶ x!	6	3! = 6 in X
÷	6720	40320 / 6 = 6720 in X
5	5	Remaining n - r = 5 (6720 lifts to Y)
↶ x!	120	5! = 120 in X
÷	56	6720 / 120 = 56 combinations!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The RPN stack handles multi-step factorial simplification effortlessly, preventing arithmetic overflow.

Core Investigation Problem:

A lottery draws 4 winning numbers from a pool of 10. How many combinations? Calculate 10C4.

Statistical Accumulation: Mean & Variance with [Σ+]

One-Variable Data Analysis & Summations • Standard: CCSS.MATH.HSS.ID.A.2

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Roll precision dice to generate discrete data points for [Σ+] accumulation: 4, 6, 7, 8, 10. Prediction: Sum = 4 + 6 + 7 + 8 + 10 = 35. Sample size n = 5. Mean x-bar = 35 / 5 = 7.0.

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

4 Σ+ 6 Σ+ 7 Σ+ 8 Σ+ 1 0 Σ+

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
4 Σ+	1	First score entered: sample size n = 1
6 Σ+	2	Second score: n = 2
7 Σ+	3	Third score: n = 3
8 Σ+	4	Fourth score: n = 4
1 0 Σ+	5	Fifth score: n = 5 total data points

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Forgetting to clear the statistical registers ([CLΣ]) before entering a new dataset with [Σ+].

Hands-On Block Remediation: Notice that pressing [Σ+] displays the sample count n in Register X; if the first data point shows n > 1, press [CLΣ] first so prior data does not corrupt the mean.

Misconception 2: Dividing by r! and then multiplying by (n − r)! when computing C(n, r) = n! / (r!(n − r)!).

Hands-On Block Remediation: Compute n! into Y, compute r! and (n − r)! and multiply them together in X ([×]), then press [÷] once to divide n! by the combined denominator.

4. EXTENSION MATHEMATICAL INSIGHT

The [Σ+] key accumulates sum(x), sum(x^2), and count n into dedicated background statistical registers.

STUDENT LAB WORKSHEET • UNIT 11 (SC-32)

Combinatorics, Probability & Series Accumulation

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: COMBINATIONS, PERMUTATIONS & STACK FACTORIALS

- Factorial [f:] [x!] multiplies n × (n-1) × ... × 1.
- Permutations (order matters): nPr = n! / (n - r)!.
- Combinations (subsets/groups): nCr = n! / (r! × (n - r)!).

Key Sequence: 8 [↶] [x!] 3 [↶] [x!] [÷] 5 [↶] [x!] [÷]

Problem 1: Calculate 6! (6 factorial) on the calculator.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 2: How many ways can 5 runners finish 1st, 2nd, and 3rd? (5P3 = 5! / 2!).

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 3: Calculate 7C2 (combinations of 2 from 7).

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 4: Accumulate 5 and 15 with [Σ+]. What is the mean?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Unit 11: Combinatorics, Probability & Series Accumulation — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 11–12 / AP

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: A lottery draws 4 winning numbers from a pool of 10. How many combinations? Calculate 10C4.

Core Solution: 210 combinations. Keystrokes: [1][0] [f:][x!] [4] [f:][x!] [÷] [6] [f:][x!] [÷]

Core Check Keys: 1 0 ↶ x! 4 ↶ x! ÷ 6 ↶ x! ÷

Extension Challenge: Accumulate data points 12, 15, 18, 21 using [Σ+]. What is the sample count n and mean?

Extension Solution: Count n = 4, Mean x-bar = 16.5.

Ext. Check Keys: 1 2 Σ+ 1 5 Σ+ 1 8 Σ+ 2 1 Σ+

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: Calculate 6! (6 factorial) on the calculator.

Expected Answer: 720

SC-32 Verification Keys: 6 ↶ x!

Problem 2: How many ways can 5 runners finish 1st, 2nd, and 3rd? (5P3 = 5! / 2!).

Expected Answer: 60

SC-32 Verification Keys: 5 ↶ x! 2 ↶ x! ÷

Problem 3: Calculate 7C2 (combinations of 2 from 7).

Expected Answer: 21

SC-32 Verification Keys: 7 ↶ x! 2 ↶ x! ÷ 5 ↶ x! ÷

Problem 4: Accumulate 5 and 15 with [Σ+]. What is the mean?

Expected Answer: Mean = 10

SC-32 Verification Keys: 5 Σ+ 1 5 Σ+ ↷ $\bar{x},\bar{y}$

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.